| | | |
|---|---|---|
| $p$ | Mersenne exponent | 7 |
| $M_p = 2^p - 1$ | base | 127 |
| $n = p + 2$ | row number | 9 |
| $x = (p + 1)/2$ | position of the middle | 4 |
| $k$ | position in the row, from 0 | 0 … 9 |
| $d$ | distance from the middle | 0 … 4 |
| $C(n, k) = n!/(k!(n - k)!)$ | k-th entry of row n | $C(9, 3) = 84$ |
| $Cat_x = C(2x, x)/(x + 1)$ | Catalan number | $Cat_4 = 14$ |

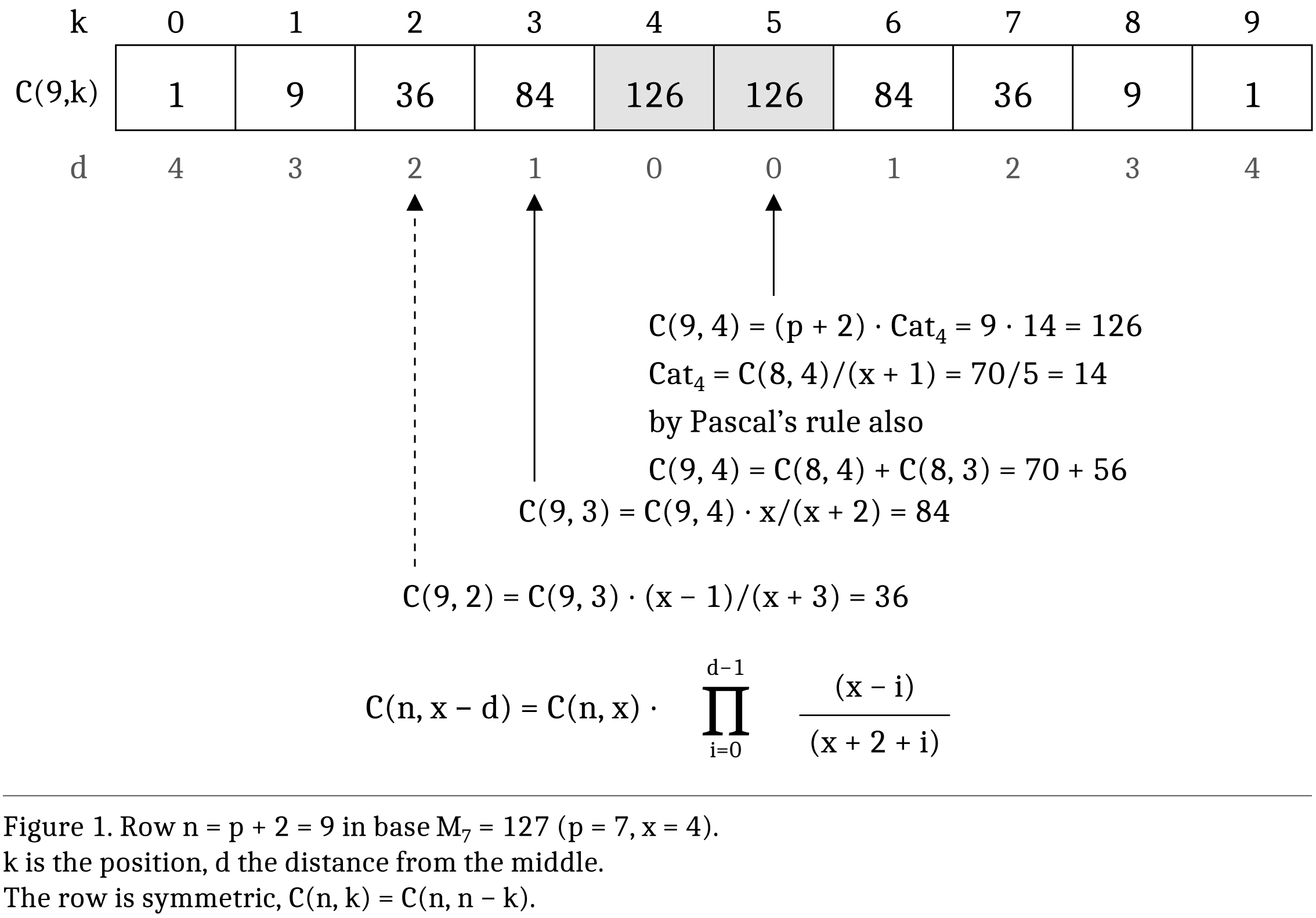


Figure 1. Row $n = p + 2 = 9$ in base $M_7 = 127$ ($p = 7$, $x = 4$).
k is the position, d the distance from the middle.
The row is symmetric, $C(n, k) = C(n, n - k)$.